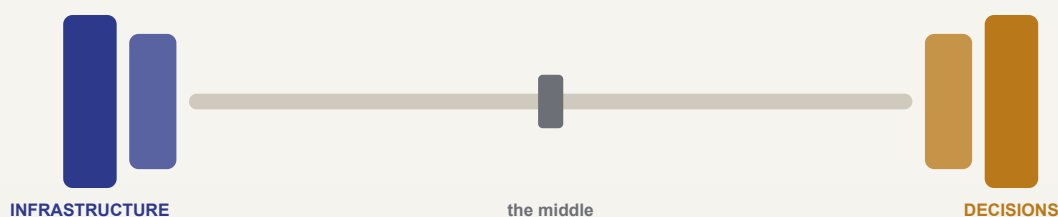


The Barbell & the *Vanishing* Middle

Why the moat in intelligent document processing is leaving extraction — and value is bifurcating into infrastructure on one side and decisions on the other.

By Anuj Sadani

“AI wrote faster and better than I could. I decided which sentences were true.”



A short narration with a book attached. Read the Overture in five minutes — or read the whole thing for the evidence, the diagrams, and the argument behind the verdict.

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The Barbell and the Vanishing Middle — an essay on where value is migrating in document intelligence, and what gets hollowed out.

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Extraction Was the Product. Now It's a Prompt.

Read only this and you will have the whole argument. Everything after it is evidence.

OPEN

Not long ago, pulling a number off a page was a business in itself. Companies existed to read the merchant name on a receipt, the gross pay on a W-2, the balance on a bank statement, and they sold accuracy by the field — the harder the field, the messier the scan, the more it was worth.

That business has not died. Pulling the *right* value, and being able to vouch for it, is still hard and still valuable. What has changed is where the difficulty — and the money — now lives. The routine nine-tenths of extraction, the part that once took a model and a team, has fallen close to a prompt. The stubborn last tenth, and the confidence to stand behind it, has not. **Extraction used to be the whole product. Today the easy part of it is a prompt, and the value has begun moving to the two ends.**

THE COLLAPSE

The classic pipeline ran in a straight line: scan the page, classify the document, extract the fields, validate the result. Four stages, and the value lived dead center — in extraction. That center is collapsing into something thin. The work didn't disappear. The *product* did.

And as the middle thins, value rushes outward to the two ends. Picture a barbell. All the weight is on the plates; the bar between them is just a bar.

THE LEFT PLATE — INFRASTRUCTURE

On one end, a new kind of company is forming. Its job is not to tell you the merchant is Walmart. Its job is to turn a two-hundred-page document into something a machine can *reason over* — clean structure, reading order, tables, figures, cross-page references, citations back to the pixel. The output is not a field. The output is a **navigable representation of the document itself.**

The tempting story is that this layer is generic, therefore cheap. Be careful. Databases are generic too. So are operating systems. Generic is how Oracle, Snowflake, and Databricks became some of the most valuable companies on earth — because *everyone builds on top of them*. If one company becomes the canonical representation layer for enterprise documents, that is not a commodity. That is a kingdom.

THE RIGHT PLATE — DECISIONS

On the other end is where the money actually is. Because no customer ever wanted extraction. They wanted to *approve the loan, deny the claim, onboard the customer, catch the fraud*. Extraction was only ever a means to an end.

A mortgage lender does not care about OCR. A mortgage lender cares about one sentence: *can I close loans faster without taking on more risk?* The closer a system gets to that sentence, the larger the budget that opens up. Decisions carry compliance, accountability, audit trails, liability. That is why they are sticky, and that is why the model providers cannot easily swallow them.

THE TRAP IN THE THESIS

Here is where the easy version of this story goes wrong. People say: the middle is becoming *simple*. It is not. **The middle is becoming invisible — and invisible is not the same as simple.**

Think of a loan package: a W-2, a paystub, a bank statement, an appraisal, a title report, a sales contract. To answer *should this loan be approved*, someone still has to resolve which “John A. Smith” across six documents is the same legal person, reconcile figures that should agree, notice the ones that don’t, reason about dates, estimate confidence, and keep a trail of where every fact came from. That is the middle. It did not vanish. It became a **load-bearing wall** — sealed inside the house, invisible from the street, and absolutely holding up the roof.

The hard problem was never “find the merchant name.” The hard problem was always: *does this name, appearing in seventeen places, refer to one legal entity — and does that entity satisfy policy XYZ?* Mortgage is not document reading. Insurance is not document reading. Trade finance is not document reading. They are **domain reasoning problems wearing the costume of document problems.**

THE VERDICT

So here is the claim, sharpened until it's hard to attack:

IDP is no longer differentiated by extraction alone. The moat is migrating outward — to document-infrastructure companies on the left and decision-automation companies on the right — while the extraction-and-reasoning layer, including its genuinely hard last mile of trustworthy, calibrated output, becomes embedded inside both ends rather than sold as a standalone product.

THE THESIS, 75% RIGHT — AND THE 25% WORTH ARGUING

The original instinct — value is leaving extraction — is correct. The correction is only this: the middle isn't dying. It's being absorbed into both ends and losing its name. Read on for the evidence.

The Pipeline That Ate Itself

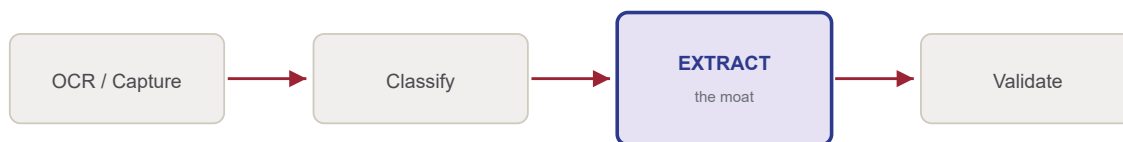
For most of its life, intelligent document processing was a story about the middle. Open almost any vendor's "ultimate guide" from the last few years and you find the same canonical assembly line: ingestion, preprocessing, classification, **data extraction**, post-processing, validation, routing, and a feedback loop that learns from human corrections.¹ Every one of those guides positions the same stage as the heart of the system — the moment raw pixels become a field with a name and a value.^{1,2}

This was not an accident of marketing. It was where the difficulty genuinely lived. Legacy optical character recognition read a page the way a photocopier does: it mapped characters in sequence without any sense of how they related to their neighbors.⁸ It relied on rigid coordinate templates that shattered the moment a vendor moved a logo or a form gained a line.⁶ So the entire art of the field was template engineering — building and maintaining brittle rules that could survive contact with the real, messy world of documents.

Product differentiation followed the difficulty. The competitive question in IDP, for a decade, was narrow and sharp:

"Who can extract harder fields, from messier documents, at higher accuracy?"

Whoever answered that question best won the deal. Accuracy on field extraction was the moat, the demo, and the invoice line item. The two ends of the pipeline — getting documents in, and pushing decisions out — were treated as bookends. Necessary, but not where the magic was. The magic was in the middle.



A straight line, with the value pooled dead center.

Figure 1. The classical IDP pipeline. For a decade, the third box was the product.

The trouble with building your business on the middle of a straight line is that straight lines are easy to re-route once a more capable component shows up. And in 2023, a more capable component showed up.

The Great Compression

The arrival of capable multimodal language models did something specific and irreversible: it turned the hardest part of the old pipeline into a single instruction. Where a team once trained, tuned, and babysat a template model to pull a tax registration number off an invoice, an engineer can now write a sentence describing the field and a schema describing its shape. The field falls out the other end. **Extraction, the thing that was once the entire product, became a line in a prompt.**

You can see the compression in the demos that now circulate as ordinary. One widely shared prototype processes forty-seven loan applications in forty-five seconds — doing extraction, cross-checking documents against each other, scoring risk, and rendering a decision — in roughly three hundred lines of code.¹⁵ A decade ago, the extraction step alone would have been a procurement cycle and a six-figure contract. Now it is a weekend.

THE SHAPE OF THE SHIFT

**60–70% →
90%+**

Straight-through (no-human) processing rates, traditional vs. agentic document systems.¹⁴

**2 months →
1–2 weeks**

Decision-cycle time on complex contracts after re-architecting around LLM-native pipelines.⁷

~94%

RAG question-answering accuracy achievable when document *preparation* is done well — the value moving left.¹⁴

It would be easy — and wrong — to read these numbers as “the problem is solved.” What actually happened is subtler. The *difficulty* did not evaporate; it changed address. The old difficulty was template design and rule maintenance. The new difficulty is orchestration: enforcing structured output, scoring confidence, catching the model when it hallucinates a plausible-but-false value, and wiring feedback loops that improve the system over time.^{8,1}

This is the first place the naive thesis stumbles. “Extraction is a prompt” is true as a statement about *cost* and *packaging*. It is false as a statement about *rigor*. A model that is mostly right is not a smaller version of a model that is right — in a regulated workflow it is a liability, because someone downstream will act on the wrong number. As the chief executive of one extraction vendor put it with unusual bluntness:

“*Mostly correct is a costly failure.*”

Be precise about what actually got cheap. For clean, common documents, accuracy on the routine fields crossed into the high nineties years ago — that part genuinely collapsed into a prompt. What did *not* collapse is the last mile: the messy scan, the adversarial layout, the rare field, and above all **calibrated confidence** — a system knowing when it is wrong. Raw accuracy is even document-dependent: clean print is near-perfect, while dense tables and handwriting are not. Pulling the right value *and being able to underwrite it* remains a hard, high-end business, and the debate over machine certainty is nowhere near settled. That residue — small in volume, large in consequence — is exactly the part that does not commoditize. It gets absorbed.

So the compression is real, but it is a compression of *price and visibility*, not of substance. The field that used to cost a contract now costs a token. What that does — what it must do — is push the locus of value away from the field itself and toward the two things a field cannot give you on its own: a *representation* rich enough to reason over, and a *decision* worth paying for. Left, and right.

*When the center of a thing becomes free, the value does not
disappear.
It migrates to the edges.*

Documents Become a Database

Watch where the most interesting infrastructure companies are pointing and you notice they have stopped trying to answer domain questions. They are not competing to tell you the merchant is Walmart. They are competing to produce something more fundamental: a faithful, machine-navigable rendering of an arbitrary document — a packet split into its constituent files, each one classified, page and section boundaries detected, layout understood, reading order recovered, tables reconstructed down to the cell, checkboxes read as checked or unchecked, signatures and stamps located, figures captioned, cross-page references resolved, and every claim traceable back to its position on the original page.^{3,4}

Notice how little of that is “extraction” in the old sense. Almost none of it is pulling a named field. It is *document understanding* — the unglamorous structural work of turning a stack of pixels into an object with parts: documents, pages, sections, tables, cells, fields, marks, signatures. That structural layer is the real left-side product. Get it wrong and everything downstream — every entity, every reconciliation, every decision — inherits the error.

The industry has a name for the inflection now: the move from *legacy OCR* to *agentic document processing*. The defining sentence in the better analyses is telling — enterprises “don’t want flat text files, they need AI-ready data (structured JSON, clean Markdown, and semantic insights) that can power LLMs and autonomous agents.”³ The deliverable has changed. It is no longer a field. It is a substrate.

The winning question is no longer “can I extract field X?” It is “can any future AI system reason over this document efficiently?”

That is a different problem, and it is a deeper one. Tools in this layer increasingly converge on the same ambition from different angles: LlamaParse reconstructs nested tables and multi-column hierarchies into clean Markdown; Unstructured.io partitions dozens of file formats for private-cloud pipelines; ColPali skips text entirely and maps page *images* to vectors for layout-aware retrieval; Memgraph’s toolkit turns raw text into a knowledge graph of nodes and edges ready for traversal.^{3,6} Different bets, one destination: the document as a queryable object.

The “generic, therefore cheap” mistake

Here the easy thesis stumbles a second time. The reasoning goes: this layer is horizontal, everyone needs it, horizontal things commoditize, therefore the left is a low-margin race to the

bottom. The commoditization pressure is real — when many tools emit interchangeable JSON-and-Markdown, buyers shop on price and developer experience.³ But “generic” and “cheap” are not synonyms, and the history of computing is a graveyard of people who confused them.

Consider the most generic software categories ever invented. A relational database stores *anything* — it knows nothing about your business. An operating system runs *any* program. Maximum genericity. And yet:

GENERIC DID NOT MEAN CHEAP

Oracle

Built a multi-decade empire on a product that stores rows it doesn't understand.

Snowflake

One of the largest software IPOs in history — for a place to put data.

Databricks

A generation-defining valuation, for a substrate everyone else builds on.

Why did genericity create rather than destroy value here? Because **everyone builds on top of them**. The moment a representation becomes the thing other systems assume, switching costs compound, ecosystems form, and the “commodity” quietly becomes a tollbooth. A database is generic the way a foundation is generic — invisible, unglamorous, and the reason the building stands.

The prize on the left, then, is not “cheap parsing.” It is to become **the canonical representation layer for enterprise documents** — the format every downstream agent, every retrieval system, every compliance tool silently assumes. That is not a commodity outcome. That is the document world's equivalent of becoming the database. Cloud providers clearly see it; Textract, Document AI, and Form Recognizer are not loss leaders, they are land grabs.¹⁴

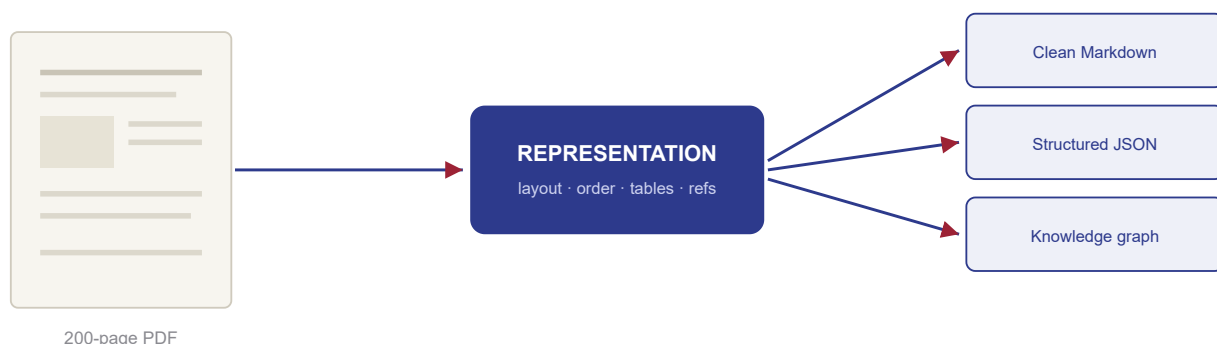


Figure 2. The left-side product is not a field — it is a navigable representation other systems build on.

IV

CHAPTER FOUR · SHIFT RIGHT

Where the Money Decides

Now walk to the other plate of the barbell, because this is where the budget actually lives. Strip away the technology and ask what a customer was ever really buying, and you arrive at an uncomfortable truth for anyone selling extraction: **no one ever wanted it**. They wanted to approve a loan, deny a claim, onboard a customer, investigate a fraud ring, file a compliance report, price an underwriting risk. Extraction was scaffolding. Nobody pays to admire scaffolding.

This is why budgets swell as you move right. A mortgage lender will not get excited about your OCR accuracy. They will get out their checkbook for one sentence:

“Can I close loans faster without increasing risk?”

THE ONLY QUESTION THAT ACTUALLY MOVES A BUDGET

The closer a system sits to that question, the more money becomes available to it — because the value being created is no longer measured in fields-per-second but in basis points of risk, days of cycle time, and dollars of liability avoided. The industry data tracks this gravitational pull. Spend and differentiation are migrating, in analyst language, “to the orchestration and platform level rather than standalone extraction modules.”¹⁴ The extraction box, increasingly, is a free or embedded feature; the decision is the product.

THE BUDGET MOVES TOWARD THE DECISION

73%

Measured ROI in an independent economic study of a deployed IDP program — value realized at the workflow, not the field.⁸

67%

Enterprises evaluating *agentic* approaches for document work in 2025 — up from 23% in 2023.¹⁰

~80%

Reduction in document-heavy transaction processing time once decisions are automated end-to-end.⁷

And once a system makes a real decision, it inherits a set of obligations that have nothing to do with reading a page — and everything to do with why this layer is defensible. A decision must be explainable: when an agent approves a claim, someone will eventually ask *why*, and the system must trace that answer back to evidence on a specific page.⁴ A decision must be auditable, governed, and accountable, because a regulator can demand the trail and a court can demand the logic.

These obligations are exactly what foundation-model providers *cannot* easily absorb. A better base model does not give you a sanctions-screening policy, a state-by-state lending rulebook, a claims-adjudication workflow, or the legal accountability for getting them wrong. That domain machinery — business rules, compliance, workflow, auditability — is what makes the right side **sticky**. It is the part of the stack that does not commoditize when the next model ships, which is precisely why the value pools there.

Extraction is a means. The decision is the end.

Budgets follow ends, not means.

The Vanishing Middle

Now to the heart of the correction — the single place where the easy version of this thesis is most wrong, and where getting it right makes the argument bulletproof. The easy version says: *the middle is becoming simple*. It is not. The middle is becoming **invisible**. Those are different claims, and the difference is the whole game.

Take a real loan package. It often arrives as a single 200-page PDF, and the very first task is unglamorous: split it into its constituent documents, detect where each one begins and ends, and classify every page — a W-2, a paystub, a bank statement, an appraisal, a title report, a sales contract. Even that is non-trivial when a packet is mis-ordered or a document spills across a boundary. And it is only the start, because the package is a small archive that disagrees with itself: each document was produced by a different party, on a different day, in a different format, for a different purpose. To answer the deceptively simple question — *should this loan be approved?* — a system must do work that no amount of clean Markdown on the left and no slick decision UI on the right will do for it:

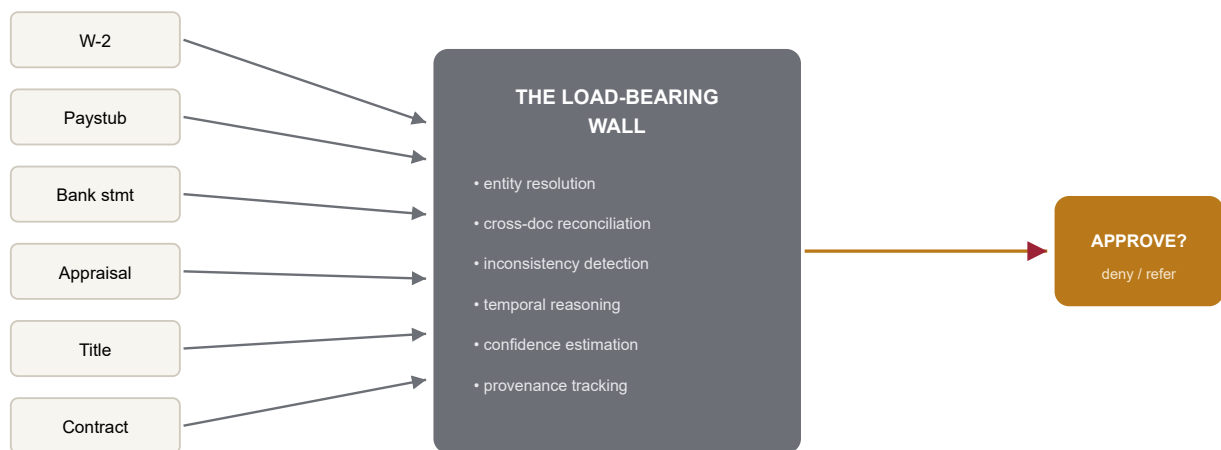


Figure 3. The middle didn't disappear. It moved inside the walls — invisible, and holding up the decision.

It must perform **entity resolution** — decide whether the “J. Smith” on the paystub, the “John A. Smith” on the title, and the “John Smith” on the contract are one legal person or three. It must do **cross-document reconciliation** — the income on the W-2 should be consistent with the deposits on the bank statement, and when they aren't, that gap is the entire signal. It needs **temporal reasoning** (is this paystub current or stale?), **confidence estimation** (how sure are we, and is that

enough to act?), and **provenance tracking** (when the regulator asks, every number points back to its source).

None of that is “reading.” All of it is the middle. And here is the point that makes the thesis hard to attack:

The middle hasn’t disappeared. It’s just no longer exposed as a standalone product. The reasoning layer still needs it — it has simply been absorbed into the walls of the systems on either side.

This is the load-bearing wall. In a finished house you do not see it. There is no plaque on the wall reading “structural — do not remove.” A visitor walking through would swear the house is just rooms and a roof. But take that wall out because it looked decorative, and the roof comes down. The middle of IDP is now that wall: sealed behind the infrastructure layer on one side and the decision layer on the other, invisible to the buyer, and absolutely holding up the result.

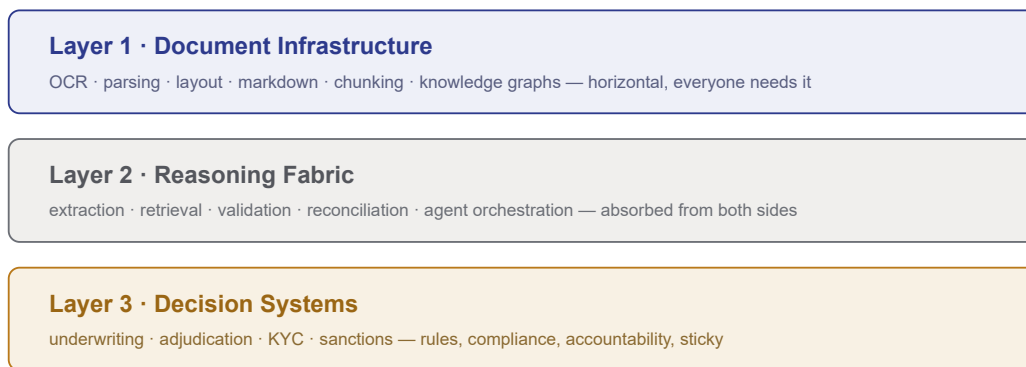
The research bears this out. Far from vanishing, the middle has grown *more* complex — it just stopped being a product you could point at. Where the old middle was hard because of templates and rules, the new middle is hard because of schema enforcement, validation, confidence scoring, multi-agent orchestration, and the deterministic harnesses now wrapped around agents to keep them from looping or hallucinating their way into a wrong decision.^{8,5} Whole tool categories — Pydantic for schema validation (over 300 million downloads a month), Instructor and BAML for forcing models into valid structured output — exist *only* to make this invisible middle trustworthy.^{23,26} That is not the toolkit of a layer that is becoming simple.

*“LLMs can read documents now, so extraction is dead.”
But mortgage was never document reading. It was domain
reasoning
wearing a document’s clothes.*

This is the deepest error the simple thesis makes, and worth stating plainly: the hard part was never *finding* the merchant name. The hard part was always deciding whether a name appearing in seventeen places refers to one legal entity, and whether that entity satisfies policy XYZ.⁴ That problem is not solved by a model that reads better. It is solved by domain reasoning — and domain reasoning is exactly the thing that survives.

The Three Economic Layers

Step back far enough and the barbell resolves into a cleaner picture: the industry is splitting into three economic layers, each with its own physics of value. Naming them precisely matters, because most companies are standing in the most dangerous one without realizing it.



Margins compress at the top, concentrate at the bottom — with a squeeze in the middle.

Figure 4. Three layers, three different economic fates.

Layer 1 — Document Infrastructure

OCR, parsing, layout, markdown, chunking, knowledge graphs. Highly horizontal; everyone needs it. Margins on any single capability compress over time — but, as Chapter Three argued, the company that becomes the *canonical* representation does not compress, it consolidates. This layer eventually *becomes* infrastructure, in the full Oracle-and-Snowflake sense of the word: low-drama, high-gravity, and very hard to dislodge once everyone builds on you.

Layer 2 — The Reasoning Fabric

This is the layer people forget, and the one most companies are quietly standing in: extraction, retrieval, validation, reconciliation, evidence gathering, agent orchestration. It is where most “agent frameworks” live. And it is the most precarious place to plant a flag — not because the work is unimportant (it is the load-bearing wall) but because **the foundation models keep absorbing pieces of it from below, while the decision systems absorb pieces of it from above.** Squeezed from both directions, much of this layer survives as a feature inside other people’s

products rather than as a business of its own. Many companies that think they are building a platform here are really building a feature someone else will ship for free.

Layer 3 — Decision Systems

Mortgage underwriting, claims adjudication, trade-finance approval, sanctions investigation, KYC. This layer stays sticky because of what it contains beyond any model: business rules, compliance regimes, workflows, accountability, auditability. These are the things a foundation-model provider cannot commoditize with a better checkpoint, because they are not capabilities — they are *responsibilities*. This is where business value durably lives.^{11,13}

READING THE THREE LAYERS AS AN INVESTOR WOULD

Layer	Economic character	Primary risk
Infrastructure	Horizontal; winner-takes-most if you become the standard	Per-feature commoditization; cloud incumbents
Reasoning fabric	Embedded; value captured by neighbors	Absorbed by models below and decisions above
Decision systems	Vertical; sticky; high willingness-to-pay	Slow sales cycles; regulatory weight; build-vs-buy

The strategic reading writes itself. Durable companies will sit firmly on one of the two ends — or will be vertically integrated enough to collapse all three layers into a single domain-optimized product. The dangerous position is to be a pure-play in Layer 2, selling the fabric as though it were a destination, while the ground erodes from both sides.

The 25% Overstated

A thesis you cannot argue against is not strong; it is merely unexamined. So here is the honest counter-case — the quarter of the argument worth holding loosely, the places where reality is messier than a clean barbell drawing.

The middle can stall the whole machine

If you believe the middle is “invisible,” you can talk yourself into believing it is handled. It often isn’t. Errors in extraction and reconciliation propagate silently into decisions, and a great many real IDP projects stall not on ingestion or workflow but on accuracy in exactly this layer.⁹ Invisible is not the same as solved. A load-bearing wall you have stopped inspecting is how roofs come down.

LLM extraction is not free of failure modes

The same models that compressed extraction into a prompt also brought new and stranger failure modes. Engineers report language models looping, hallucinating values, or tripping content filters mid-page; structured parsing of dense tables can be *harder* for a model than for an old rule-based extractor unless carefully engineered.¹⁴ Counter-intuitively, adding chain-of-thought reasoning has been shown to *hurt* raw parsing accuracy while raising latency and hallucination.¹⁴ “Just prompt it” is a real capability and a real liability at once.

Paper has not read the memo

The clean two-layer future assumes documents arrive as files. A large share — by some surveys around 61% of document processes — still involves paper somewhere in the chain.¹⁴ Paper-heavy, legacy-bound industries are exactly the high-value verticals (legal, finance, government), and they migrate slowly and expensively. The barbell is the destination; the journey is longer and lumpier than a diagram admits.

The graph-versus-retrieval question is genuinely open

Part of the “documents become a database” story leans on knowledge graphs as the rich representation. The evidence is mixed: some studies find multi-agent GraphRAG decisively beating plain vector retrieval on multi-hop reasoning; others find naive GraphRAG doing *worse* than simple, well-tuned dense retrieval.¹⁴ Enterprises that over-invest in heavyweight graph

machinery may end up outperformed by a boring, well-engineered search index. The left layer's winning *form* is not settled.

And the lines genuinely blur

Finally, the cleanest critique of any layer cake is that the real market refuses to stay in its boxes. The large suites — UiPath's document-plus-agent platform, Tungsten's TotalAgility spanning standard IDP through multi-agent workflows and RAG-based knowledge discovery, the end-to-end stacks from AWS, Azure, and Google — deliberately weave all three layers into one product, so the buyer never perceives a middle to begin with.^{11,14} That convergence is itself evidence for the thesis (the middle disappears as a category) and a caution against it (no clean left/right split survives contact with a platform sales motion).

None of this overturns the core claim. It sands down its edges. Value *is* migrating to the ends; the middle *is* losing its standing as a product. But the migration is slower, the middle is more fragile-yet-essential, and the boundaries are blurrier than a confident one-line prediction lets on. Hold the thesis at 75%, not 100%, and it gets stronger, not weaker.

The Verdict

So where is intelligent document processing actually going? Not, as the easy version claims, toward “no more extraction.” Extraction is still being rebuilt — on agentic OCR, on layout grounding, on vision-language models — and its trustworthy last mile is still a real, hard business. What is ending is extraction as the *moat*: the place where differentiation concentrates and budgets are won. The real motion is outward — toward the two heavy ends of the barbell — and inward, as the old center is swallowed by the walls on either side.

Stated as a prediction for 2026 through 2028, sharpened until it is hard to attack:

IDP is no longer differentiated by extraction alone. The moat is migrating outward — to document-infrastructure companies on the left and decision-automation companies on the right — while the extraction-and-reasoning layer, including its genuinely hard last mile of trustworthy, calibrated output, becomes embedded inside both ends rather than sold as a standalone product category.

THE VERDICT

Notice what that sentence does *not* say. It does not say the middle is simple — it is a load-bearing wall, more intricate now than it ever was, merely sealed out of view. It does not say the left is cheap — the company that becomes the canonical representation layer for enterprise documents will be the document world’s Snowflake, not its commodity. It does not say extraction stopped mattering — only that it stopped being a thing you sell, and became a thing you assume, like a database query or a network call.

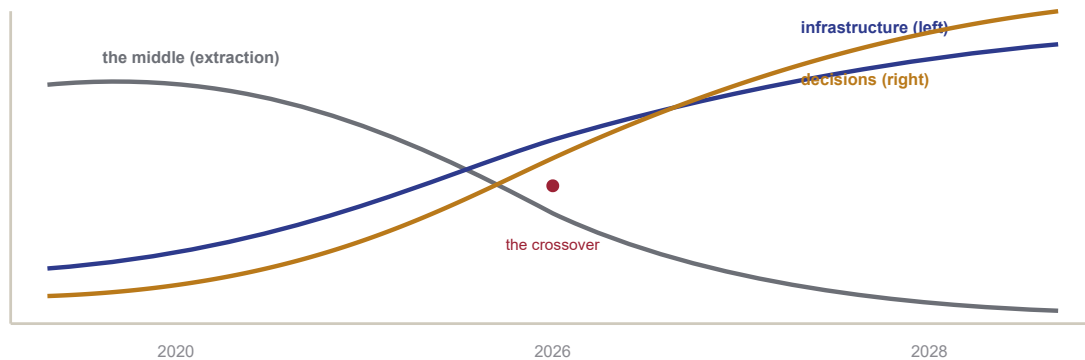


Figure 5. Value migration, 2020–2028: the middle peaks and recedes as the two ends climb past it.

What to do about it

If you are building: pick an end, or integrate hard enough to own all three layers in one vertical. On the left, do not aim to parse — aim to become the representation everyone assumes. On the right, do not aim to extract — aim to own the decision, with its rules, its compliance, and its accountability. The one position to avoid is a pure-play in the reasoning fabric, selling the load-bearing wall as if buyers shopped for walls. They don’t. They buy houses.

If you are buying: capture value at both ends. Invest in genuinely good ingestion and representation, and invest in agentic decision automation on workflows you can govern — but never assume the invisible middle is handled. Inspect the wall. “Mostly correct” is still a costly failure, and it fails most quietly precisely where you have stopped looking.

The first instinct — that value is leaving extraction — was right. The correction is the whole point: the middle is not dying. It is being absorbed into both ends and losing its name. The barbell is real. The bar between the plates is still there, still bearing load. You just can’t see it anymore.

*The routine nine-tenths of extraction became a prompt.
The trustworthy last mile became a wall you cannot see —
holding up everything that pays.*

Building for 2028

It is one thing to say value moves to the ends. It is another to build a company there. So let us do the useful exercise: zoom out to 2028 and ask what the product actually looks like — the stack, the org chart, the things each team does on a Tuesday — and then zoom back in to the parts that are near-certain, so the vision stays honest. The goal is not a roadmap. It is a map of where you would innovate and where you would simply execute.

The stack in 2028

The three economic layers are still there, but each has hardened into recognizable sub-layers — the kind of thing a buyer will expect to see on an architecture slide. Note where the old “middle” lives: not as a product of its own, but as named capabilities sealed inside the fabric.

Layer 1 — Document Infrastructure

horizontal · becomes-the-standard · the document's database

Connectors & capture

Splitting & boundary detection

Classification & routing

Agentic OCR / VLM

Layout & reading order

Table structure (cell-level)

Form & checkbox state

Signature & stamp detection

Figure & chart capture

Document graph

Provenance & citation index

Embeddings & retrieval

Representation store

Layer 2 — Reasoning Fabric (*the load-bearing wall*)

invisible as a product · absorbed from both sides · where trust is manufactured

Schema-as-config extraction

Entity resolution

Cross-document reconciliation

Temporal reasoning

Confidence calibration

Evidence assembly

Agent orchestration

Eval & guardrail harness

Layer 3 — Decision Systems

vertical · sticky · where the budget and the liability live

Policy & rules engine

Case & workflow management

Human-in-the-loop review

Explainability & audit trail

Compliance & governance

Outcome analytics & feedback

Accountability ledger

Who builds it, who sells it

If extraction is no longer the moat, the whole organization re-points around the *decision*. Here is what that means for each function in 2028 — concrete, not aspirational.

SALES

Sells the decision, not the document. Lands a vertical with an **outcome contract** — basis points of risk, days off the cycle, dollars of liability avoided. The headline is a decision SLA, never an accuracy number.

MARKETING

Tells a **trust story**, not a feature story. Shows before/after cycle times, audited decisions, “close faster without more risk.” Proof is reference customers and third-party evals — not an OCR percentage on a slide.

PRODUCT / PM

Owns the decision and its **exception path**. Sets confidence thresholds and escalation policy, curates the eval sets, and prioritizes by dollars-per-decision and risk-per-error rather than features shipped.

ENGINEERING

Builds representation, fabric, and decision logic as **one system**. Wraps agents in deterministic harnesses, integrates to systems of record, and treats the eval suite — not unit tests alone — as the real definition of done.

DEVOPS / LLMOPS

Runs model routing and versioning, **regression evals on every change**, drift and hallucination monitoring, cost/latency budgets, prompt and schema versioning, data residency, and fast rollback. Treats a prompt change like a deploy.

OPERATIONS / SUPPORT

Runs the **human-in-the-loop**: exception queues, reviewer tooling, and feedback labeling that feeds the evals. Incident response is decision response — in this business, a bug is a wrong decision someone acted on.

The frontier and the bedrock

This is the zoom-out and the zoom-in, side by side. The left column is where you earn a moat by 2028 — the unsolved, high-leverage problems. The right column is what is near-certain regardless of who wins, the ground you build on rather than bet against.

FRONTIER — WHERE YOU INNOVATE

- **Confidence you can underwrite** — calibrated, per-decision certainty a risk officer will sign their name to.
- **Self-improving reconciliation** — fabric that learns each vertical's characteristic inconsistencies instead of being hand-tuned.
- **Cross-organization document graphs** — one shared representation spanning counterparties, not a silo per company.
- **Accountable agents** — provable, replayable audit trails for autonomous decisions a regulator will accept.

BEDROCK — NEAR-CERTAIN BY 2028

- The routine ~90% of extraction is commoditized and effectively free.
- The trustworthy last mile is still hard — and still where errors hurt most.
- Humans still own the long tail and the liability; full autonomy stays the exception.
- Regulation has hardened: audit and explainability are table stakes, not features.
- Buyers purchase outcomes and platforms, not standalone extraction modules.

Read the two columns together and the strategy falls out on its own. You innovate on the left column because that is where a defensible moat is still unclaimed; you do *not* waste capital fighting the right column, because those are settled facts you build on top of. The companies that win in 2028 will be the ones that picked an end of the barbell, treated the bedrock as given, and spent every scarce dollar of cleverness on the frontier — on manufacturing trust inside the wall nobody can see.

*Zoom out far enough and everything looks like agents.
Zoom back in and the work is the same as it ever was:
be right, know when you're not, and be able to prove it.*

APPENDIX

Notes & Sources

Figures and quotations are drawn from the supporting research compiled for this essay (vendor documentation, analyst commentary, and peer-reviewed studies, accessed 2026). Citation numbers in the text map to the sources below.

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About the author

Anuj Sadani builds AI systems and the teams that build them. He spent a decade at NVIDIA, has worked inside AI-first organizations, and has led multicultural engineering pods across Europe — sixteen-plus years, most of them on the AI era's unglamorous half: turning hype into systems that ship, and systems that ship into outcomes that hold up in production. That work has fed into industry recognition from Gartner and Everest Group, and an Innovator of the Year nod.

He is the author of *The Clean Vibe Coder: A Code of Conduct for Programmers in the Age of AI Agents* and *Borrow the Line. Own the Move.*, and writes about engineering, leadership, and what stays human when the tools get good. He still believes the best technology is the kind that makes the people around it braver.



Writing and work

Essays, books, and what he is building.

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